DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (DATA SCIENCE)
ABES ENGINEERING COLLEGE, GHAZIABAD

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project Title: Retail Inventory Management System

SUBMITTED BY	UNDER THE GUIDANCE OF	DATE OF SUBMISSION
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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a complete, clear, and unambiguous definition of the functionality, external interfaces, performance targets, and constraints of the proposed system: 'Retail Inventory Management System'. This specification serves as a binding reference agreement between the student developer, project evaluators, the academic supervisor, and prospective end users during the 3rd-year engineering curriculum.

1.2 Scope of the System

Retail businesses that operate from several warehouses must always know what stock they hold, where it is stored, what has already been promised to customers and what must be reordered. When this information is kept in spreadsheets, registers or disconnected tools, it becomes outdated quickly and different people see different numbers. The proposed application solves this problem by delivering an end-to-end web platform, called the Retail Command Center, backed by a single relational database. Key boundary guidelines include:

- **In-Scope Capabilities:** Centralized product, warehouse, inventory, order, supplier and purchase-order data; user registration and JWT-based authentication; role-based workflows for admin, supplier and customer; automatic stock reservation and low-stock alerts; notifications; dashboard visualization and CSV reports.
- **Out-of-Scope Boundaries:** Dedicated mobile applications, AI-based demand forecasting, external ERP / accounting integrations, and live commercial payment or shipping-carrier integration.
- **Expected Benefits:** A single trusted source of stock information, fewer manual data-entry errors, faster access to stock and order status, timely reorder alerts, and immediate accessibility from any modern browser.

1.3 Definitions, Acronyms, and Abbreviations

Category	Term / Acronym	Definition / Standard Context
Standard	**SRS**	Software Requirements Specification (IEEE Std 830-1998 format).
Architecture	**API**	Application Programming Interface for data interchange via JSON/HTTP.
Security	**JWT / RBAC**	JSON Web Token for stateless authorization; Role-Based Access Control.
Security	**BCrypt**	Salted one-way hashing algorithm used to store passwords.
Database	**CRUD**	Create, Read, Update, Delete basic data manipulation operations.
Database	**JPA / ORM**	Java Persistence API; maps Java entity classes to PostgreSQL tables.
Network	**REST**	Representational State Transfer stateless software architectural style.
Project Specific	**SKU**	Stock Keeping Unit – unique code identifying a product.

Category	Term / Acronym	Definition / Standard Context
Project Specific	**PO**	Purchase Order – an order placed with a supplier to replenish stock.
Project Specific	**Reserved stock**	Quantity set aside for a placed order but not yet shipped.
Project Specific	**Reorder level**	Minimum available quantity of a product below which a low-stock alert is raised.

1.4 Document Overview

The remainder of this SRS is partitioned into structural sections: Section 2 outlines the product perspective, user classes, operating environment, constraints and the Agile development approach. Section 3 itemizes detailed functional requirements organized into the eleven software modules, with an explanation of how each module works. Section 4 specifies external interfaces (UI, Hardware, Software, Communication). Section 5 defines non-functional attributes. Section 6 provides the academic modeling diagrams. Section 7 documents the test cases, including a detailed authentication test plan, and Section 8 holds the review sign-off.

2. Overall Description

2.1 Product Perspective

The project is structured as a self-contained multi-tier system comprising a presentation layer (React + TypeScript), an application/API logic tier (Spring Boot REST services secured with Spring Security and JWT) and a persistent PostgreSQL database accessed through Spring Data JPA. It interfaces with an external SMTP mail server, used only for password-reset e-mails, through the Spring mail client.

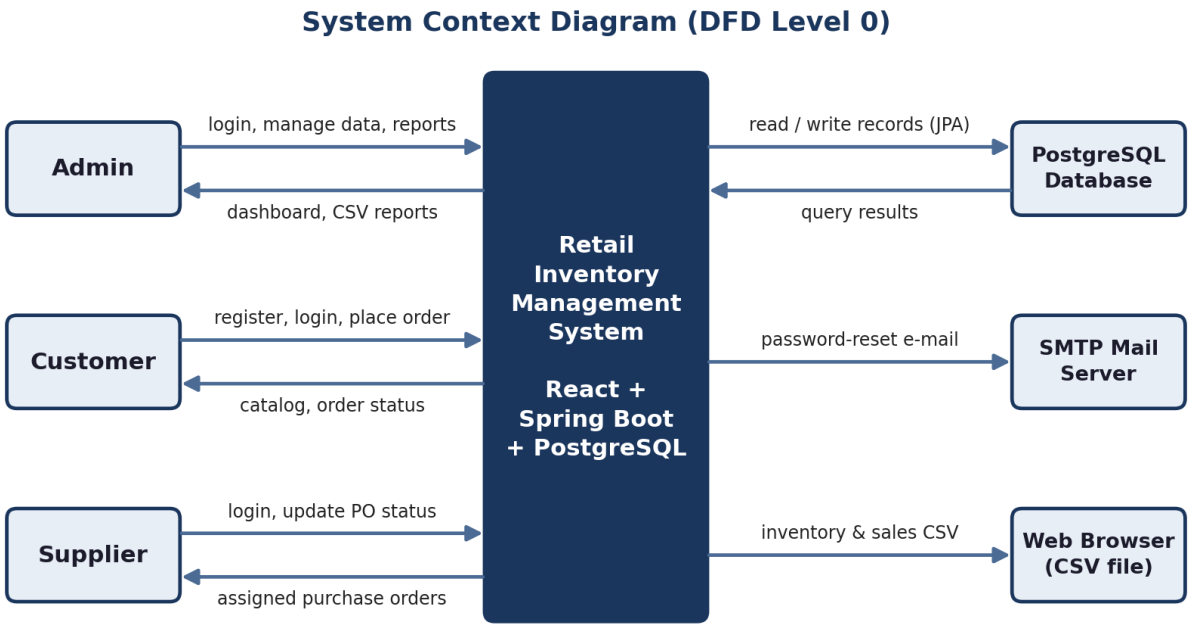


Figure 2.1: System context diagram (DFD Level 0)

2.2 User Classes and Characteristics

The software accommodates distinct user personas with varying access hierarchies:

User Role / Persona	Technical Proficiency	Core Responsibilities & Rights
System Administrator (ROLE_ADMIN)	High (operations / data management)	Complete access: dashboard, products, inventory, orders, fulfillment, suppliers, purchase orders, reports and user management, including suspending or activating accounts.
Supplier (ROLE_SUPPLIER)	Basic–Intermediate	Logs in to the supplier portal, views purchase orders assigned to the supplier, updates their status and edits item details.
Customer (ROLE_CUSTOMER)	Basic (general web user)	Registers and logs in, browses the product catalog by category, adds items to the cart, places orders and tracks own order history.
Registered User (ROLE_USER)	Basic	Base role of every account; authenticated profile management and catalog access.
Guest / Visitor	Basic	Can only reach the login and registration page; every other page and API requires authentication.

2.3 Operating Environment

The application operates within the following software and hardware environments:

- **Client Requirements:** Modern web browser (Google Chrome, Microsoft Edge, Mozilla Firefox, Safari); minimum display resolution 1366x768 for desktop.
- **Server Runtime Environment:** Java 17 (LTS) with Spring Boot 3.2 on Windows, Linux or macOS; Node.js LTS is required only to build and run the Vite frontend. The backend runs on port 8080 and the frontend development server on port 5173.
- **Database Storage Tier:** PostgreSQL (database rism_db) accessed through the PostgreSQL JDBC driver; optionally run with Docker Compose.

2.4 Design and Implementation Constraints

- **Academic Budget Constraint:** Only free / open-source tools are used; the system runs locally or in Docker containers without paid cloud services.
- **Security Compliance:** Passwords are stored only as BCrypt hashes; the API is stateless and every request except /api/auth/** requires a valid JWT; tokens expire after 24 hours.
- **Modularity Standards:** Layered architecture (Controller → Service → Repository → Entity) on the backend and page/component/service separation on the frontend ensures clean separation of concerns.

2.5 Assumptions and Dependencies

- **Assumptions:** Users have a working Internet / network connection and a modern browser; PostgreSQL is running and reachable before the backend starts.
- **Dependencies:** A configured SMTP account (for password-reset e-mails) and the open-source libraries declared in pom.xml and package.json.

2.6 Development Methodology

The project follows an Agile approach delivered in 15 incremental sprints. Each sprint produced a working, testable increment, allowing requirements to be refined as the system grew.

Sprint(s)	Focus / Deliverable
1	Requirements, architecture, ER design, Git repository and project initialization.
2	Authentication and authorization: login, JWT, roles.
3–5	Product catalog, warehouses and basic inventory.
6–7	Inventory transactions, stock adjustments and warehouse transfers.
8–9	Order creation with stock validation and fulfillment workflow.
10–11	Suppliers, purchase orders and supplier portal.
12–13	Reports, dashboard enhancements, notifications and audit logs.
14–15	Dockerization, testing, bug fixing and deployment preparation.

3. System Features and Functional Requirements

Requirements are categorized by functional subsystem modules with unique requirement identifiers (FR-x.x). Each module is first explained in brief and then itemized.

3.1 Module 1: Authentication and Access Governance

Controls who may enter the system and what each user may do. The login page posts credentials to /api/auth/login; Spring Security compares the password with its BCrypt hash and, if valid, a JWT is returned. The frontend stores the token, loads the user's roles from /api/users/me and redirects to the correct portal.

Req ID	Feature Description	Input / Validation Rule	Priority
FR-1.1	User Account Registration	Username (unique), e-mail, password, account type (customer / supplier); duplicate username rejected	High
FR-1.2	Secure Credential Login	Username + password; verifies BCrypt hash; on failure returns "Incorrect username or password"	High
FR-1.3	Session Token Management	Signed JWT valid for 24 hours, sent as Authorization: Bearer header	High
FR-1.4	Role-Based Access and Redirection	Admin → /dashboard, Supplier → /supplier/dashboard, Customer → /customers/portal; unauthorized pages redirected	High
FR-1.5	Forgot Password	Registered e-mail receives reset link; unknown e-mail gets a neutral message	Medium
FR-1.6	Suspended Account Block	Accounts with non-active status refused with HTTP 403 message	Medium

3.2 Module 2: Dashboard

Gives administrators a quick summary of business activity on login. It reads inventory, order and purchase data from the REST APIs and shows them as summary cards and Recharts charts with links to detailed modules.

Req ID	Feature Description	Input / Validation Rule	Priority
FR-2.1	Summary Metric Cards	Aggregated counts of products, stock, orders and purchase orders	Medium
FR-2.2	Charts and Recent Activity	Order/status trends and recent orders rendered with Recharts	Low

3.3 Module 3: Product Catalog

Maintains the master list of items the business sells. Each product has a unique SKU, name, category, description, cost price, selling price, reorder level, status and image.

Req ID	Feature Description	Input / Validation Rule	Priority
FR-3.1	Add / Edit / Deactivate Product	SKU unique and mandatory; cost and selling price numeric; reorder level integer	High
FR-3.2	Category Management	Category name unique; products linked through category_id	Medium
FR-3.3	Search and Filter	Free text on product name / SKU; filter by category	High

3.4 Module 4: Inventory and Warehouses

Shows how much of each product is held in each warehouse. The inventory record links a product to a warehouse and tracks total, reserved and available quantity; warehouse records store name, location, capacity and status. Saving an inventory record re-checks the product's total available stock and raises a low-stock alert to inventory managers when it is below the reorder level.

Req ID	Feature Description	Input / Validation Rule	Priority
FR-4.1	Warehouse-wise Stock View	Total, reserved and available quantity per product and warehouse (unique pair)	High
FR-4.2	Stock Update / Adjustment	Quantity integer; available quantity cannot be negative (database CHECK)	High
FR-4.3	Warehouse Management	Name, address / city, capacity, status	Medium
FR-4.4	Stock Transfer and Transaction Records	API stores transfer (source, destination, quantity > 0) and stock transaction history records	Medium

3.5 Module 5: Orders

Captures customer orders and commits stock correctly. When an order is created, the order service goes through every item, reduces the available quantity and increases the reserved quantity across the warehouses holding that product. If the remaining available stock drops below the reorder level, a low-stock alert is sent to inventory managers.

Req ID	Feature Description	Input / Validation Rule	Priority
FR-5.1	Create Order	Customer, warehouse, items (product, quantity > 0, unit price), total amount	High
FR-5.2	Automatic Stock Reservation	Available quantity decreases and reserved quantity increases by the ordered amount	High
FR-5.3	Low-Stock Alert	Triggered when total available quantity < product reorder level	High
FR-5.4	Order Notifications	“Order Placed” to creator and “New Order” to order managers	Medium

3.6 Module 6: Fulfillment

Moves an order from placement to delivery. The fulfillment board groups orders by status (Pending, Packed, Shipped) and the admin updates them through PUT /api/orders/{id}/status. When an order becomes Shipped or Delivered, its reserved quantity is released and the total quantity is permanently reduced.

Req ID	Feature Description	Input / Validation Rule	Priority
FR-6.1	Order Status Update	Status such as PENDING, CONFIRMED, PROCESSING, PACKED, SHIPPED, DELIVERED, CANCELLED	High
FR-6.2	Inventory Finalization	On SHIPPED / DELIVERED: reserved and total quantity reduced	High
FR-6.3	Status-Change Notification	Order creator notified with the new status	Medium

3.7 Module 7: Suppliers and Purchase Orders

Keeps a register of vendors (company, contact person, e-mail, phone, address, status) and manages replenishment. A purchase order is created for a supplier with product lines (quantity, unit cost, total cost) and moves through statuses such as Pending, Accepted, Shipped and Received. Supplier users receive a “New Purchase Order” notification, and procurement managers are notified when the supplier accepts it.

Req ID	Feature Description	Input / Validation Rule	Priority
FR-7.1	Supplier Management	Add, edit, view and delete supplier records	High
FR-7.2	Create Purchase Order	Supplier, items with quantity > 0 and unit cost; total amount	High

Req ID	Feature Description	Input / Validation Rule	Priority
FR-7.3	Purchase Order Status Tracking	Status updates; PO Accepted notifies procurement managers	Medium

3.8 Module 8: Supplier Portal

Lets suppliers handle their own purchase orders without admin rights. After login, a ROLE_SUPPLIER user sees only the supplier dashboard and purchase orders assigned to them, served by the /api/supplier endpoints.

Req ID	Feature Description	Input / Validation Rule	Priority
FR-8.1	View Assigned Purchase Orders	Orders filtered by the logged-in supplier	High
FR-8.2	Update Order Status	Status change request on a purchase order	High
FR-8.3	Edit Item Details	Update item information of an assigned order	Medium

3.9 Module 9: Customer Portal

Provides an online shop front. Customers browse products by category or collection, add items to a cart, place orders and view their own order history with expandable order details. They cannot open admin pages.

Req ID	Feature Description	Input / Validation Rule	Priority
FR-9.1	Browse and Filter Catalog	Category / collection filters	Medium
FR-9.2	Cart and Place Order	Product and quantity added to cart, then submitted as an order	High
FR-9.3	Order History	Only the logged-in customer’s orders and status shown	Medium

3.10 Module 10: Users and Profile

Administers accounts and lets users maintain their own details. Password hashes are never returned by the API.

Req ID	Feature Description	Input / Validation Rule	Priority
FR-10.1	List Users (Admin)	All users displayed without password hashes	Medium
FR-10.2	Activate / Suspend Account	Status field toggled by admin; suspended user cannot log in	Medium
FR-10.3	Update Own Profile	Full name, e-mail, phone, address	Low

3.11 Module 11: Reports and Notifications

Supports monitoring and decision making. Two report endpoints produce downloadable CSV files, and notifications (low stock, new order, status change, purchase-order events) are stored per user and can be marked as read.

Req ID	Feature Description	Input / Validation Rule	Priority
FR-11.1	Inventory CSV Report	GET /api/reports/inventory/csv – ID, SKU, name, category, selling and cost price	Medium
FR-11.2	Sales CSV Report	GET /api/reports/sales/csv – order ID, date, status, total amount	Medium
FR-11.3	User Notifications	List per user; mark as read	Medium

4. External Interface Requirements

4.1 User Interfaces (UI)

The interface follows standard UX usability practice: a clean sidebar navigation, role-specific menus, modal dialogs for data entry and tables with status badges. All forms provide immediate visual feedback (success banners or inline error alerts such as “Incorrect username or password”).

4.2 Hardware Interfaces

- **Standard Deployment:** Standard input/output peripherals (keyboard, mouse, display of at least 720p). No special hardware is required.
- **Embedded / IoT Variants:** Not applicable to this project.

4.3 Software Interfaces

- **Relational Database Connector:** Spring Data JPA / Hibernate with the PostgreSQL JDBC driver (database rism_db, port 5432).
- **Mail Service:** Spring Boot Mail using SMTP (STARTTLS on port 587) for password-reset e-mails.
- **Frontend Libraries:** Axios for REST calls, React Router for navigation, Recharts for charts.

4.4 Communication Interfaces

- **Web Traffic Protocol:** HTTP between the browser and the REST API (backend port 8080) during development; production deployment must use HTTPS with TLS.
- **Data Serialization:** All payload bodies are exchanged as UTF-8 application/json, using a common envelope with success, message and data fields; reports are returned as text/csv.

5. Non-Functional Requirements (NFRs)

Attribute	Target Metric / Standard	Verification & Testing Method
5.1 Performance	Normal pages load within about 3 seconds and API calls respond within about 1 second under typical classroom load.	Browser developer tools network tab; Postman response times.
5.2 Security	Zero plain-text passwords (BCrypt); stateless JWT with 24-hour expiry; all endpoints except /api/auth/** need a valid token; role-restricted routes; password hashes never returned by the API.	Authentication test cases TC10–TC13 and TC19; manual role-based access checks.
5.3 Reliability	Consistent storage through relational keys and CHECK constraints; automatic stock reservation inside a database transaction.	Order test cases TC30–TC34; Docker restart policy.
5.4 Portability	Works in Chrome, Edge, Firefox and Safari; runs on any OS with Java 17 and PostgreSQL.	Manual cross-browser inspection.
5.5 Usability	Role-specific navigation, clear error messages and status badges understandable by non-technical staff.	Walkthrough with sample users.
5.6 Maintainability	Modular layers (controller, service, repository, entity) and typed React components.	Code review; module-wise testing.

6. System Design and Analysis Models

University evaluation rubrics require standard graphical analysis models illustrating the dynamic and static behavior of the application. The models prepared for this project are summarized below and shown in the figures that follow.

Diagram Type	Objective & Scope	Deliverable Format
Use Case Diagram	Maps actors (Admin, Customer, Supplier) to their functional interactions.	UML use case diagram (Figure 6.1).
Data Flow Diagram (DFD)	Information flow: Level 0 (Figure 2.1) and Level 1 modular decomposition (Figure 6.2).	DFD with processes and data store.
Entity-Relationship (ERD)	Entities, primary / foreign keys and relationships of the PostgreSQL schema.	ERD with 1:N relationships (Figure 6.3).
Sequence Diagram	Message flow between UI, controller, security and database for the login feature.	UML sequence diagram (Figure 6.4).

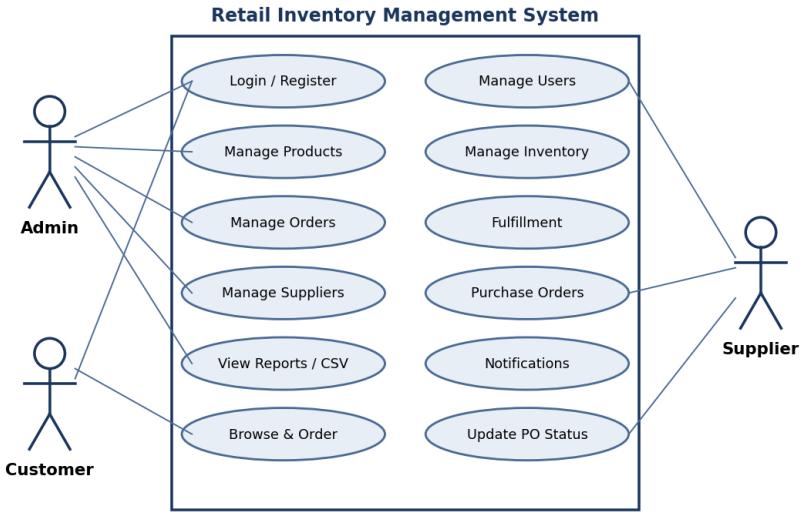


Figure 6.1: Use case diagram

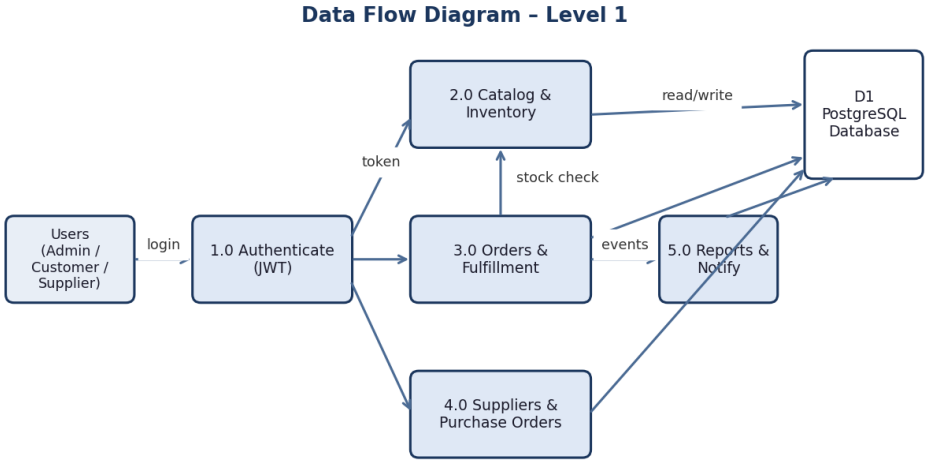


Figure 6.2: Data flow diagram – Level 1

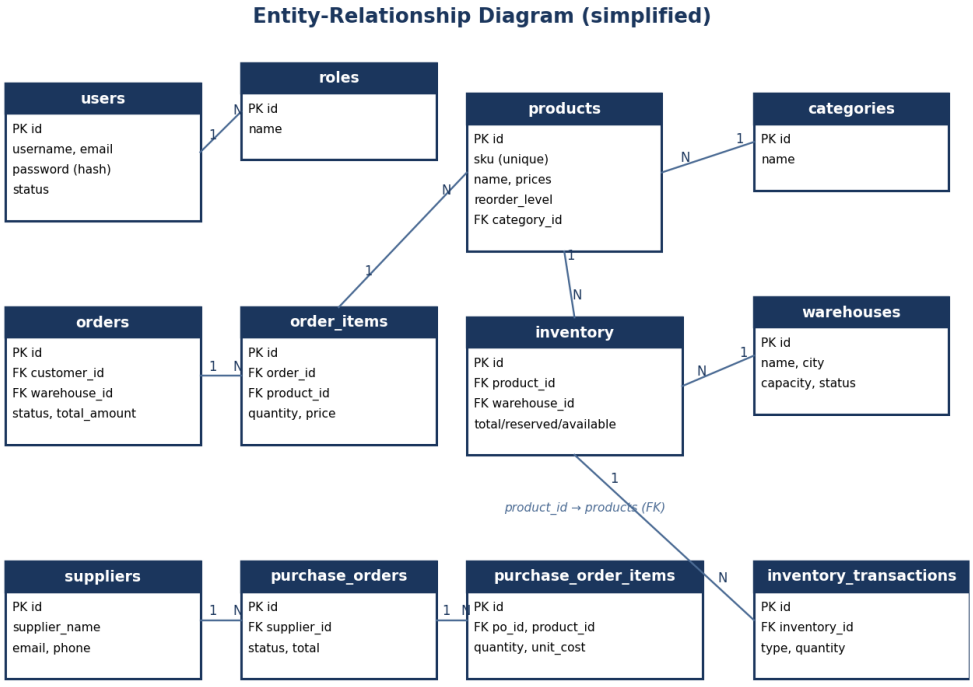


Figure 6.3: Entity-relationship diagram (simplified)

Normalization note: the schema is in third normal form – customer, product, category, warehouse and supplier data are stored once and referenced by foreign keys; order and purchase-order line items are held in separate child tables (order_items, purchase_order_items); users and roles are linked through the user_roles join table.

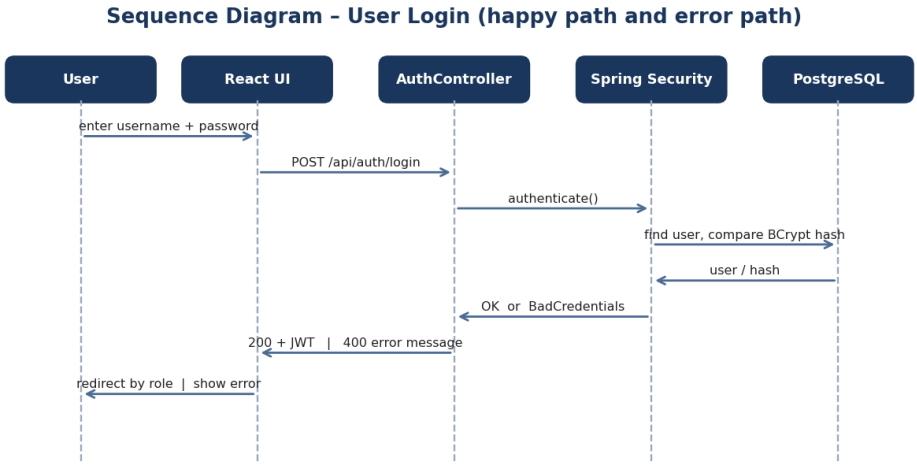


Figure 6.4: Sequence diagram – user login

6.1 Database Tables

Table	Important Fields	Purpose
users / roles / user_roles	id, username, password (hash), email, status; role name	Access and authorization
categories / products	sku, name, category_id, cost_price, selling_price, reorder_level, status	Catalog data

Table	Important Fields	Purpose
warehouses	name, address, city, capacity, status	Warehouse data
inventory	product_id, warehouse_id, total / reserved / available quantity	Warehouse-wise stock
inventory_transactions / stock_transfers	inventory_id, type, quantity; source / destination warehouse	Stock history and transfers
suppliers / purchase_orders / items	supplier_name, contact; supplier_id, status, total; product_id, quantity, unit_cost	Procurement data
customers / orders / order_items	customer_id, warehouse_id, status, total_amount; product_id, quantity, unit_price	Retail order data
shipments / notifications / audit_logs	carrier, tracking_number; type, message, is_read; action, details	Fulfillment, alerts, traceability

7. Testing and Validation

Testing is planned at functional level for every module through the user interface and the REST endpoints. Each test case lists the input and the expected result. Authentication is covered in the greatest detail because every other module depends on it. The Status column is to be updated after executing each case.

7.1 Authentication Test Cases

Endpoints under test: POST /api/auth/login, POST /api/auth/register, POST /api/auth/forgot-password and the /login page.

ID	Test	Input	Expected Result	Status
TC01	Valid user login	Registered, active username and correct password	Login successful; JWT token issued; user redirected by role (Admin → Dashboard, Supplier → Supplier Dashboard, Customer → Customer Portal).	Pending
TC02	Invalid password	Valid username with wrong password	Access rejected (HTTP 400); “Incorrect username or password” shown; no token issued.	Pending
TC03	Invalid username	Unregistered username with any password	Access rejected; “Incorrect username or password” shown; no token issued.	Pending
TC04	Invalid username and password	Both username and password incorrect	Access rejected; “Incorrect username or password” shown; no token issued.	Pending
TC05	Empty fields	Submit login form with blank username / password	Browser validation stops submission; no request is sent.	Pending
TC06	Suspended account	Correct credentials of an account suspended by admin	Access rejected (HTTP 403); “Restricted by admin. Your account has been suspended.” shown.	Pending
TC07	Admin role redirect	Login with an admin account	Redirected to /dashboard; all admin menu items visible.	Pending

ID	Test	Input	Expected Result	Status
TC08	Supplier role redirect	Login with a supplier account	Redirected to /supplier/dashboard.	Pending
TC09	Customer role redirect	Login with a customer account	Redirected to /customers/portal.	Pending
TC10	Protected page without login	Open /dashboard without logging in	User is redirected to the login page.	Pending
TC11	Role restriction	Customer or supplier opens an admin-only URL such as /users	User is redirected to their own portal; admin page not shown.	Pending
TC12	API call without token	Call a protected endpoint (e.g. GET /api/products) without Authorization header	Request rejected; no data returned.	Pending
TC13	Invalid or tampered token	Send a modified / expired JWT in the Authorization header	Request rejected; user must log in again.	Pending
TC14	Logout	Click logout, then open a protected URL	Session cleared; user is redirected to the login page.	Pending
TC15	Register new account	Unique username, valid e-mail, password and account type	Account created; success message shown; user can then sign in.	Pending
TC16	Duplicate username	Register with an existing username	Rejected with “Username already exists”.	Pending
TC17	Forgot password – registered e-mail	Enter a registered e-mail	Success message; reset link e-mail sent (requires SMTP setup).	Pending
TC18	Forgot password – unknown e-mail	Enter an unregistered e-mail	Neutral message “If that email exists, a reset link was sent.” – account existence not revealed.	Pending
TC19	Password storage	Inspect the users table after registration	Password is stored as a BCrypt hash, not plain text.	Pending

7.2 Product and Inventory Test Cases

ID	Test	Expected Result	Status
TC20	Product search	Matching product / SKU records displayed.	Pending
TC21	Category filter	Only products of the selected category are shown.	Pending
TC22	Add product	Product stored and shown in catalog.	Pending
TC23	Duplicate SKU	Rejected by the unique constraint; error returned.	Pending
TC24	Edit product	Updated price / details are saved and displayed.	Pending
TC25	Inventory view	Warehouse-wise total, reserved and available quantities load.	Pending

ID	Test	Expected Result	Status
TC26	Update inventory below reorder level	Inventory saved; low-stock notification created for inventory managers.	Pending
TC27	Add warehouse	New warehouse saved and listed.	Pending
TC28	Stock transfer record	POST /api/stocktransfers stores the transfer with source, destination and quantity.	Pending

7.3 Order and Fulfillment Test Cases

ID	Test	Expected Result	Status
TC29	Create order	Order saved; “Order Placed” notification generated for the creator.	Pending
TC30	Stock reservation	Available quantity decreases and reserved quantity increases by the ordered amount.	Pending
TC31	Low-stock alert on order	When remaining stock falls below reorder level, a low-stock notification is created.	Pending
TC32	Update order status	New status saved and the order creator notified of the change.	Pending
TC33	Ship / deliver order	Reserved quantity released and total quantity reduced.	Pending
TC34	Customer order history	Customer sees only their own orders with details.	Pending
TC35	Add to cart	Selected product and quantity appear in the cart.	Pending

7.4 Procurement and Supplier Test Cases

ID	Test	Expected Result	Status
TC36	Add supplier	Supplier record saved and listed.	Pending
TC37	Edit / delete supplier	Changes are reflected in the supplier list.	Pending
TC38	Create purchase order	PO saved with items, total cost and pending status; supplier users notified.	Pending
TC39	Supplier views POs	Supplier sees only purchase orders assigned to them.	Pending
TC40	Supplier accepts PO	Status updated to ACCEPTED; procurement managers notified.	Pending
TC41	Mark PO received	PO status changes to RECEIVED and is displayed in the procurement list.	Pending

7.5 Reports, Users and Other Test Cases

ID	Test	Expected Result	Status
TC42	Dashboard	Summary cards and charts load with current data.	Pending

ID	Test	Expected Result	Status
TC43	Inventory CSV report	GET /api/reports/inventory/csv downloads inventory_report.csv.	Pending
TC44	Sales CSV report	GET /api/reports/sales/csv downloads sales_report.csv.	Pending
TC45	View users (admin)	User list displayed without password hashes.	Pending
TC46	Suspend user	Account status changes; that user can no longer log in (see TC06).	Pending
TC47	Update profile	Name, e-mail, phone and address are saved.	Pending
TC48	Notifications	Notifications appear for the user and can be marked as read.	Pending

8. Academic Review & Sign-Off

This Software Requirements Specification document has been submitted and reviewed by the academic supervisory panel:

Project Coordinator / Guide (Avinay Raj)	Internal Examiner	Head of Department (HOD)
Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____